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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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shall not indicates an interdiction (prohibition) to do something

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The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

Introduction

The present document is part of a TS family covering the 3rd Generation Partnership Project Technical Specification Group Services and System Aspects, Management and orchestration; as identified below:

TS 28.314: "Plug and Connect; Concepts and requirements".

TS 28.315: "Plug and Connect; Procedure flows".

TS 28.316: "Plug and Connect; Data formats".

1 Scope

The present document specifies concepts, use cases and requirements for *Plug and Connect* NE in 3GPP systems.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.310: "Network Domain Security (NDS); Authentication Framework (AF)".

[3] IETF RFC9810: "Internet X.509 Public Key Infrastructure Certificate Management Protocol".

[4] IETF RFC 4211: "Internet X.509 Public Key Infrastructure Certificate Request Message Format (CRMF)".

[5] 3GPP TS 38.401: "Technical Specification Group Radio Access Network; NG-RAN; Architecture description".

[6] 3GPP TS 28.532: "Technical Specification; Management and orchestration; Generic management services".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

Plug and Connect: The procedure by which a NE gets basic connectivity information after it is powered up and gets connected to its management system.

Software and Configuration Server (SCS): A server that provides software and configuration functions for each connected network element.

IAB-node: as defined in TS 38.401 [5].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

CA	Certification Authority
CMP	Certificate Management Protocol
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
FQDN	Fully Qualified Domain Name
NAT	Network Address Translation
NE	Network Element
PnC	Plug and Connect
RA	Registration Authority
SCS	Software and Configuration Server
SeGW	Security Gateway
VLAN	Virtual LAN
IAB	Integrated Access and Backhaul

4 Concepts and background

4.1 Plug and Connect Concept

4.1.1 General description

Plug and connect is a list of procedures for connecting the NE to its management system. The basic steps of Plug and Connect are described in clause 6.1.1.

The entities involved in the PnC concept are NE, DHCP server, DNS Server, Certification Authority server, SCS (including the Initial and Serving SCS that could be the same in certain deployment scenarios), Security Gateway.

4.1.2 Network Scenarios

4.1.2.1 NE connected via a Non-Secure, Operator Controlled Network

An NE is typically connected to the operator's network according to one of the following scenarios:

In Figure 4.1.2.1.1, the NE is connected directly to a network controlled by the operator. The NE can use IP Infrastructure services (DHCP Server, DNS Server, etc.) in the Non-secure Operator Network. The Operator has full control of these nodes. One or more Security Gateways protect the Secure Operator Network from malicious NEs. Within the Secure Operator Network, there are also IP Infrastructure nodes.

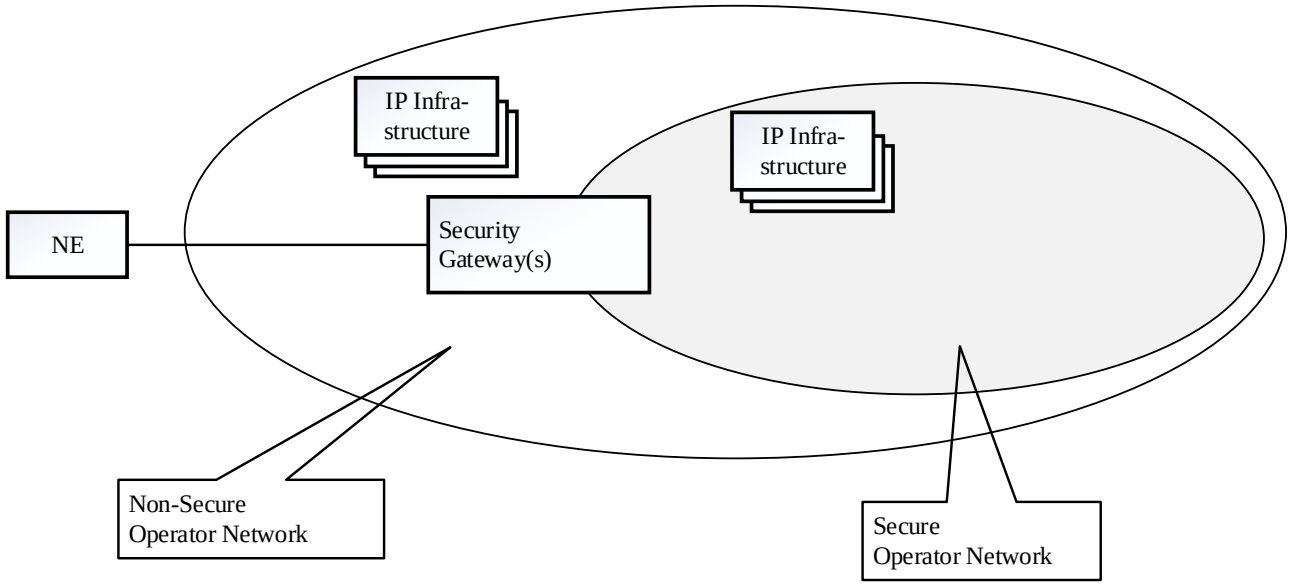


Figure 4.1.2.1.1: NE connected to a Non-Secure Operator Network

4.1.2.2 NE connected via an External Network

In Figure 4.1.2.2.1, the NE is connected to a network controlled by an entity external to the Operator. In contrast to the first scenario, the IP Infrastructure nodes in the External Network are not fully controlled by the operator. In both cases, the NE needs to traverse the Security Gateway(s) to access the nodes in the Secure Operator Network.

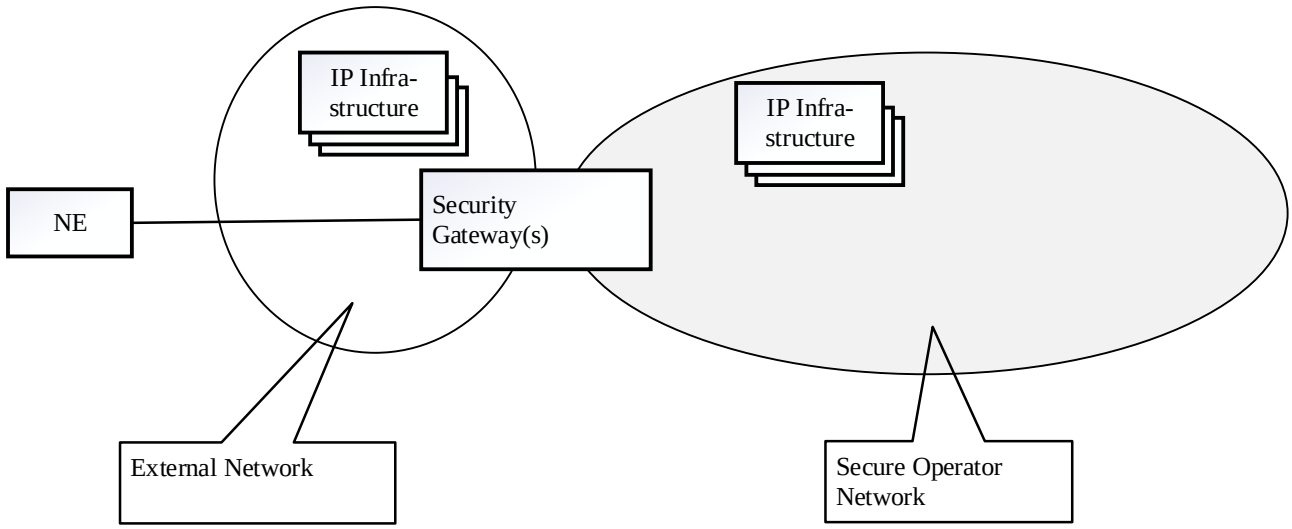


Figure 4.1.2.2.1: NE connected to an External Network

4.1.3 Security Aspects

4.1.3.1 Root Certificate Acquisition:

In accordance to TS 33.310 [2] clause 9.2 there are two options how to obtain the operator root certificate:

Option 1: The operator root certificate is provisioned in the NE prior to the CMPv2 protocol run.

Option 2: The operator root certificate is provisioned in the NE during the CMPv2 protocol run (as part of the Initialisation Response).

The required pre-provisioning in option 1 is against the basic idea of PnC to minimize pre-provisioning. Therefore from the PnC perspective Option 2 is more interesting. From a security point of view the following considerations are relevant:

- Option 2 has the risk that during the CMP initialisation a man-in the middle attack could take place. In order to be successful, such an attack happens timely during the actual CMP initialization run and the attacker has access to the access network between NE and RA/CA.
  This risk can be assessed as acceptable, given (a) the risks which are present at Options 1's prior provisioning – see below, (b) the short time window of vulnerability, (c) the closed access networks of many operators. In addition, most attacks will only lead to inability of the NE to connect to the network, or to misuse of the new NE by the attacker. The operator should notice it soon if the NE does not connect and will investigate the issue.

- Option 1 avoids the above "time window of vulnerability". On the other hand, it requires pre-provisioning of the operator root certificate, either in factory or on-site by service personnel. There is the risk of a security leak during the provisioning of the root certificate within the vendor / commissioning environment.

It seems questionable from a security point of view to allow option 2 also in public Internet (without operator-trusted access network). There the attacks stated above are more probable, and an attacker may even install some (static) catching or spoofing equipment in the public Internet to always capture such "initialization requests".

It is up to the network operator to choose the option with is preferable from his point of view (risk assessment, Plug and Connect importance).

The enrolment of NE shall use the CMPv2 protocol as specified in IETF RFC 9810 [3] and IETF RFC 4211 [4]. Security mechanism is further specified in TS 33.310 [2] clause 9.3.

4.1.3.2 Number of CA servers

There could be one or more RA/CA server, e.g. one per NE vendor. If more than one RA/CA server is deployed with one RA/CA server per vendor then the vendor identification would be needed either in the FQDN of the RA server or in the information from the IP AutoConfiguration Service carrying the information about RA/CA server.

4.1.3.3 Number of OAM SeGWs

There could be one or more OAM SeGW, e.g. one per NE vendor. If more than one OAM SeGW is deployed with one OAM SeGW per vendor then the vendor identification would be needed either in the FQDN of the OAM SeGW or in the information from the IP AutoConfiguration Service carrying the information about OAM SeGW.

5 Business Level Requirements

5.1 Business Requirements for Plug and Connect

REQ_PnC_CON_1 Plug and Connect shall use standard protocols.

REQ_PnC_CON_2 VPN tunnels needed for Plug and Connect shall be set-up automatically.

REQ_PnC_CON_3 The complete key management during Plug and Connect shall be a fully automatic secure procedure, based on procedures defined by 3GPP SA3.

REQ_PnC_CON_4: It shall be possible to perform the Plug and Connect procedures using secure protocols and procedures between the NE and OAM.

REQ_PnC_CON_5 An NE shall be able to get its own IP addresses and SCS IP address without manual configuration.

REQ_PnC_CON_6 For Plug and Connect the SCS shall only be accessible by authenticated and authorized NEs.

REQ_PnC_CON_7 For Plug and Connect the initial and final configuration of the NE (or the information how to retrieve them) shall only be accessible by authenticated and authorized NEs.

REQ_PnC_CON_8 The Plug and Connect solution shall be usable for IPv4-only networks, for IPv6-only networks and for dual stack IP networks.

REQ_PnC_CON_9 Plug and Connect procedures shall support connection of NEs with and without NAT and via External Networks or Non-Secure Operator Networks.

6 Specification Level Requirements

6.1 Use Cases

6.1.1 Use case Plug and Connect

Table 6.1.1.1

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	After physical installation, connect the NE to its SCS as automatically as possible.	
Actors and Roles	NE as user. In this use case NE is the RAN NE. Other types of NE might also be compliant and use this use case. Examples of NEs are: - gNB - eNB The NE within virtualization is not addressed.	
Telecom resources	NE; IP networks: Non-Secure Operator Network, External Network, and its elements like DHCP server optionally DNS, CA/RA servers, Security Gateway(s) (each protecting one or more Secure Operator Networks), Secure Operator Network(s) including SCS(s)	
Assumptions	There is a functional power supply for the NE. There may be one or more IP Autoconfiguration Services like DHCP and Router Advertisements and zero or more DNS servers.	
Pre conditions	The NE is physically installed. IP connectivity exists between the involved telecom resources. The involved telecom resources are functional. The relevant information is stored and available: - Vendor Certificate at the NE - Operator Certificate at the CA/RA - For the External Network or Non-Secure Operator Network: - (Outer) IP autoconfiguration information at the IP Autoconfiguration Service - FQDN of the initial OAM SeGW at the NE and/or FQDN or IP address of the initial OAM SeGW at the IP Autoconfiguration Service - FQDN of the CA/RA servers at the NE and/or FQDN or IP address of the CA/RA servers at the IP Autoconfiguration Service - If FQDNs need to be resolved, corresponding IP address(es) at the DNS server(s) - For the Secure Operator Network: - (Inner) IP autoconfiguration information at the IP Autoconfiguration Service or at the initial OAM SeGW - FQDN or IP address of the initial SCS at the NE and/or DHCP Server of the Secure Operator Network. - If FQDNs need to be resolved, corresponding IP address(es) at the DNS server(s) - Configuration and software for the NE at the SCS(s)	
Begins when	The NE is powered up.	
Step 1 (M)	If a VLAN ID is available the NE uses it. Otherwise the NE uses the native VLAN where PnC traffic is sent and received untagged	
Step 2 (M)	The NE acquires its IP address through stateful or stateless IP autoconfiguration. This may provide 0 or more DNS server addresses.	

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Step 3 (M)	The NE acquires the IP address of the CA/RA server. The FQDN of the CA/RA server may be pre-configured in the NE or the FQDN or IP address of the CA/RA server may be provided by the IP Autoconfiguration Service. FQDNs are resolved through the DNS if necessary. Information provided by the IP Autoconfiguration Services shall supersede those pre-configured at the NE.	
Step 4 (M)	The NE performs Certificate Enrolment.	
Step 5 (M)	The NE acquires the IP address of the OAM SeGW. The FQDN of the OAM SeGW may be pre-configured in the NE or the FQDN or the IP address of the OAM SeGW may be provided by the IP Autoconfiguration Service. FQDNs are resolved through the DNS if necessary.	
Step 6 (M)	The NE establishes a secure connection (tunnel) to the Security Gateway given by Step 5. The NE receives its (inner) IP autoconfiguration information (which may be the same as the outer IP address obtained in step2) and optionally the address of one or more DNS servers within the Secure Operator Network from the Configuration Parameters of IKEv2 during tunnel establishment.	
Step 7 (M)	The NE acquires the IP address of the correct Element Manager by either, issuing a DHCP request including the NE's vendor information, resolving FQDNs via DNS if necessary, or by having a pre-configured FQDN (including the NE's vendor information) resolved via DNS.	Secure connection
Step 8 (M)	The NE establishes a connection to the provided SCS and acquires its configuration and software if any. The configuration may contain an address to another SCS that this specific node shall use as SCS. The configuration may contain an address to another SeGW that should be used before connecting to the SCS. The NE may then - release the connection to the current SCS and OAM SeGW and then restart (returning to step 1), - release the connection to the current SCS and OAM SeGW and then return to step 6, - release the connection to the current SCS and then repeat step 8.	Secure connection
Ends when	Ends when all mandatory steps identified above are successfully completed or when an exception occurs.	
Exceptions	One of the steps identified above fails.	
Post Conditions	One or more secure connections exist between the NE and the SCS. Via the connection to the SCS the NE can receive further instructions to become operational and carry user traffic, e.g. the administrativeState is set to "unlocked".	
Traceability	All requirements of clauses 5.1 and 6.2.1.	

Security aspects – e.g. prevention of unauthorized network access and of fake parameters supplied to the NEs, etc. - have special importance. Security related sub-steps to establish secure connections are not shown in table 6.1.1.1. More security aspects are described in clause 4.1.3.

6.1.2 Use case IAB-node connects to management system

6.1.2.1 Overview

IAB architecture is specified in 3GPP TS 38.401[5]. The IAB-node connects to an upstream IAB-node or an IAB-donor-DU via a subset of the UE functionalities of the NR Uu interface (named IAB-MT function of IAB-node). IAB-node differs from static NE in term of its wireless connection with IAB-donor-node and its mobility behaviours. Mobile IAB can move to different geographic areas or traverse to different PLMNs that require to connect to different management system based on its location.

IAB-node connects to management system at power up and during mobility by following PnC steps as described in clause 6.1.1. At step of IP autoconfiguration, IAB-node differs from static NE in term of obtaining IP configuration for OAM connectivity. When IAB-node powers up, IAB-node may get the initial IP configuration via PDU session that setup between IAB-MT and 5GC, using this connectivity IAB-node can obtain the initial IP configuration provided by IP autoconfiguration service (e.g. DHCP server), and additionally may perform DNS query depends on the operator's deployment scenario. The management system that IAB-node connects to may change due to IAB-node mobility behaviours. To support IAB-node connecting to a management system based on its location, IAB-node should obtain IP configuration for OAM connectivity that is associated to its location via notification operations as described in clause

6.1.2.2. With the IP configuration obtained IAB-node performs the subsequent PnC procedures, satisfies security aspects including CA/RA enrolment and connectivity to SeGW before connecting to SCS.

6.1.2.2 Obtain IP configuration for OAM connectivity via notifications

3GPP management system should support IAB-node to use notification mechanism to obtain IP configuration for OAM connectivity. When notification mechanism is used, 3GPP management system shall support the mapping of IP configuration for OAM connectivity with IAB-node location, see 6.1.2.3.

When IAB-node powers up and after connecting to 3GPP management system for the 1st time, the MnS producer for IAB-node shall provide location information with IAB-node indication via notifications to 3GPP management system as specified in 3GPP TS 28.532 [6] (e.g. notifyMOIChanges) to indicate its current location. Upon the reception of notifications, 3GPP management system shall provision the IP configuration for OAM connectivity (e.g. via createMOI operation) as specified 3GPP TS 28.532 [6] based on its current location.

When IAB-node moves to a new location which may need connects to new management system, the MnS producer for IAB-node shall provide new location information via "notifyMOIChanges" or "notifyMOIAttributeValueChanges" or other notifications as specified in 3GPP TS 28.532 [6] to 3GPP management system. Upon the reception of notifications of new location, 3GPP management system shall provision the new IP configuration for OAM connectivity via "modifyMOIAttributes" or other operations as specified in 3GPP TS 28.532 [6] to the MnS producer for IAB-node based on its new location.

6.1.2.3 IP configuration for OAM connectivity with location information

The IP configuration for OAM connectivity should include one or more the following depending on the deployment scenarios:

- Basic IP configuration information (e.g. IP address, netmask, default gateway, domain name, IP address(es) of DNS servers, time servers)
- FQDN or IP address of the initial OAM SeGW
- FQDN or IP address of the CA/RA servers. If FQDNs need to be resolved, corresponding IP address(es) at the DNS server(s).
- FQDN or IP address of the initial SCS. If FQDNs need to be resolved, corresponding IP address(es) at the DNS server(s)

The location information should include one or more of the following depending on IAB-node physical location or the mobility scenarios as specified in 3GPP TS 38.401 [5] clause 8.12 and clause 8.23:

- TAC or TAI pertaining to the cells where IAB-MT is connected.
- Serving cell ID that IAB-MT is connected to.
- The gNBId of the IAB-donor-CU and PLMN info that target IAB-DU connects to.
- The gNBId of the IAB-donor-CU and PLMN info that serves the IAB-MT.

6.2 Requirements

6.2.1 Specification Requirements for Plug and Connect

REQ_PnC_FUN_1 The establishment of secure tunnels from the NE to the OAM shall support NAT traversal.

6.2.2 Specification Requirements for IAB-node connecting to management system

REQ_IAB_CON_1: 3GPP management system should have the capability to support IP configuration for connecting to management system based on IAB-node location.

REQ_IAB_CON_2: 3GPP management system should have the capability to receive notification about the location of IAB-node.

REQ_IAB_CON_3: 3GPP management system should have the capability to enable IAB-node to fulfill the security aspects of PnC procedures to connect to management system at power up and during mobility.

Annex A (informative): Graphical representation of the PnC Use Case

The NE Plug and Connect procedure, given in clause 6.1.1 are classified into two sets corresponding to those conducted at External Network (or Non-secure Operator Network) and those conducted at the Secure Operator Network. An interpretation of these procedures is depicted in figures A.1 and A.2 respectively.

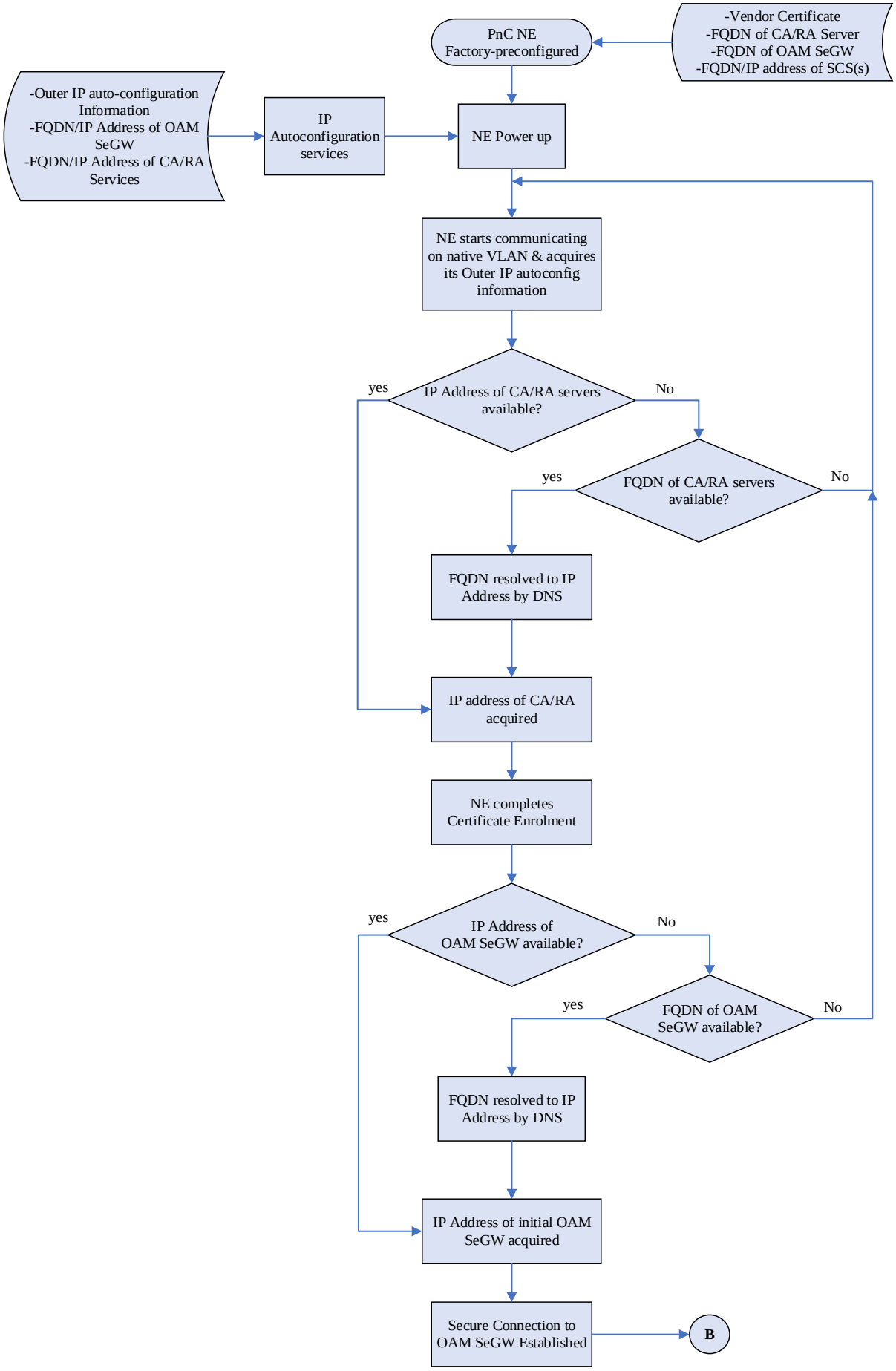


Figure A.1: PnC procedure for the External Network or Non-secure Operator Network

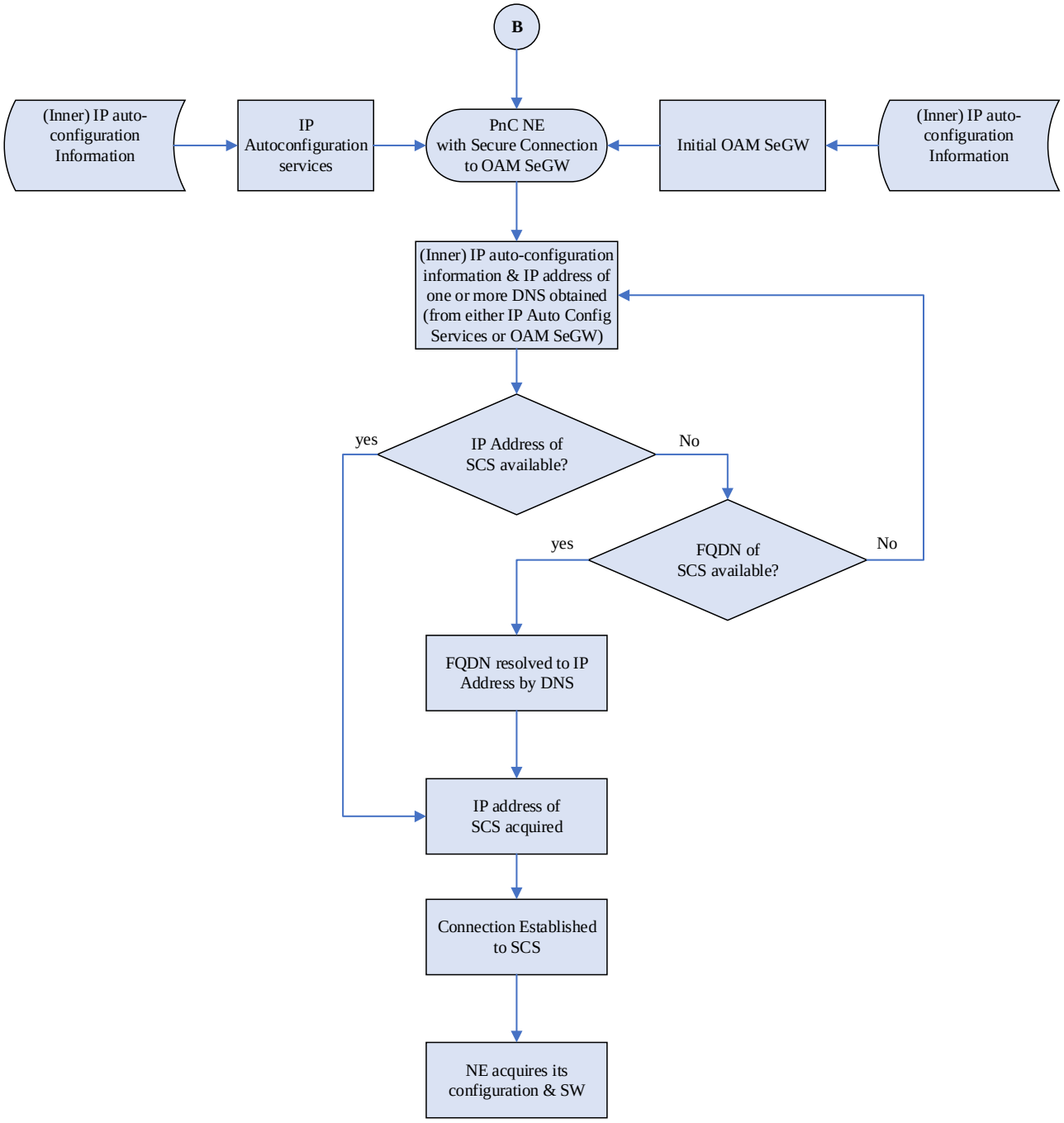


Figure A.2: PnC Procedure for the secure Operator Network

Annex B (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2021-06	**SA5#137-e**	**S5-213662**					**0.1.0**
2021-09	**SA5#138-e**	**S5-214659**					**0.2.0**
2021-10	**SA5#139-e**	**S5-215628**					**0.3.0**
2021-12	**SA5#140-e**	**S5-216602**					**0.4.0**
2022-01	**SA5#141-e**	**S5-221749**					**0.5.0**
2022-03	**SA#95e**	**SP-220122**				**Presented for information and approval**	**1.0.0**
2022-03	**SA#95e**					**Upgrade to change control version**	**17.0.0**
2024-04	**-**	**-**	**-**	**-**	**-**	**Update to Rel-18 version (MCC)**	**18.0.0**
2025-06	SA#108	SP-250543	0002	1	B	Requirements for IAB-node connects to management system	19.0.0
2025-06	SA#108	SP-250543	0004		B	Use case IAB-node connects to management system	19.0.0
2025-09	SA#109	SP-251097	0006	1	D	Editorial changes	19.1.0
2026-09	SA#113	SP-260830	0009	1	F	Rel-19 CR TS 28.314 corrections of RFC reference	19.2.0

History

Version	Date	Status
V19.1.0	October 2025	Publication
V19.2.0	October 2026	Publication